



L2B: Orders of Growth

CS1101S: Programming Methodology

Boyd Anderson

21st August, 2026

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Outline

Announcements

Tree Recursion (Textbook 1.2.2)

Orders of Growth (Textbook 1.2.3)



Calendars & Absence Policy

New Calendars

Now on Canvas: **Lectures and Events** calendar and **Timelines** calendar — subscribe to stay in sync!



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Simplified Absence Policy

Allowed without penalty, throughout the semester:

- ▶ 1 absence in Studios



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- ▶ 2 absences in Lectures



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- ▶ 2 absences in Lectures
- ▶ 1 absence in RA/PA

E.g. for Studios: we expect **10 non-zero Studio XP** out of 11 Studios for full attendance credit.



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Keep your MCs — only submit them at the end of semester if you exceed the allowance above.



Studio Attendance and Participation

- ▶ Attendance and Participation (5% of CS1101S)
 - Studios are small communities of learners; your presence in the sessions is needed to be an effective member
 - 3% for Studio attendance
 - 2% for Studio participation; Avengers assess your participation, based on effort
 - The ideal Studio member prepares well for the meetings, contributes by answering any questions thoughtfully, asks good questions, and tries to help classmates in mastering the material
- ▶ Studio Performance XP
 - Maximal 500 XP per Studio session
 - Here is where you can shine, by helping others, contributing to discussions, going the extra mile



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Fibonacci Numbers

- ▶ Fibonacci sequence
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- ▶ Our attempt to implement $F(n)$ in Python:

```
def fib(n):  
    return n if n <= 1 else fib(n - 1) + fib(n - 2)
```





Evaluating fib(5)

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fib(5)



Evaluating fib(5)

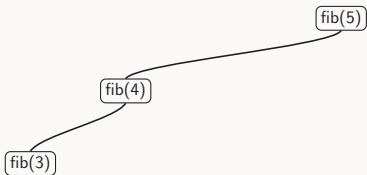
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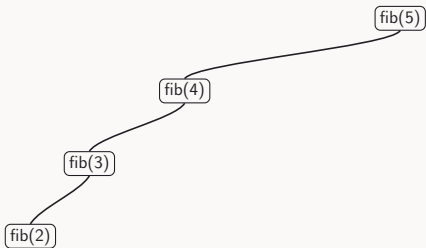
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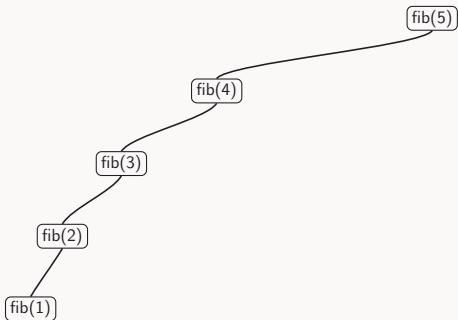
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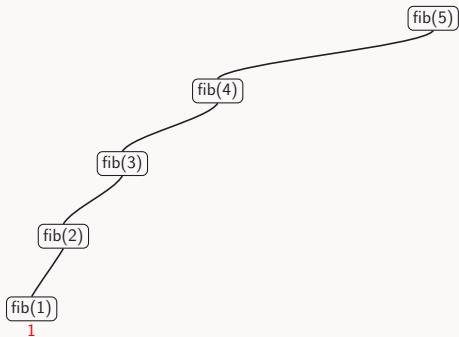
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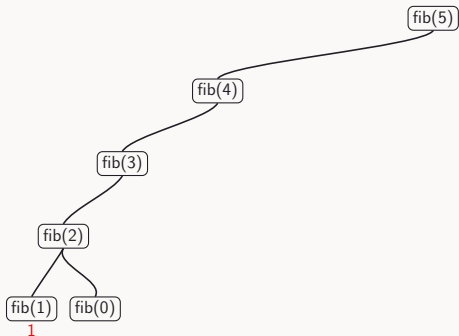
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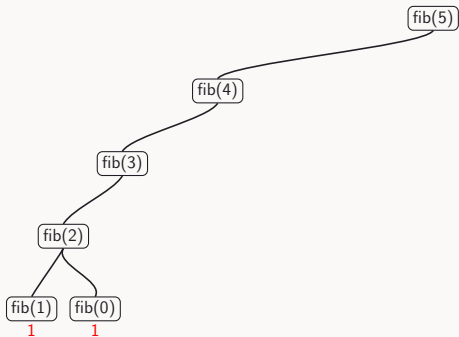
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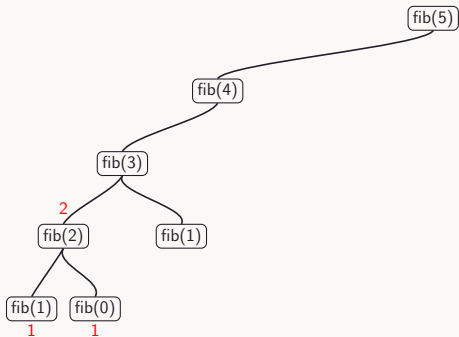
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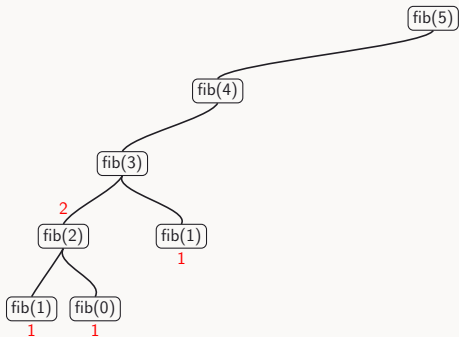
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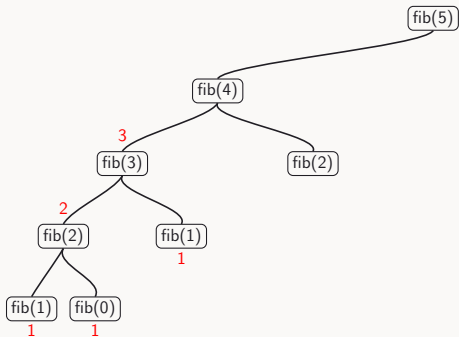
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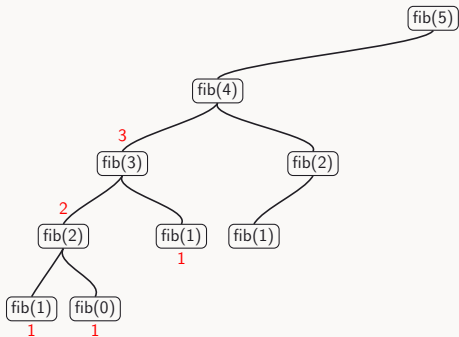
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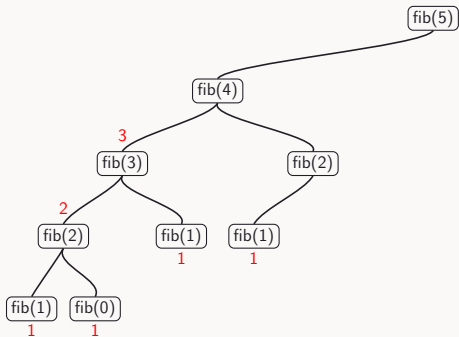
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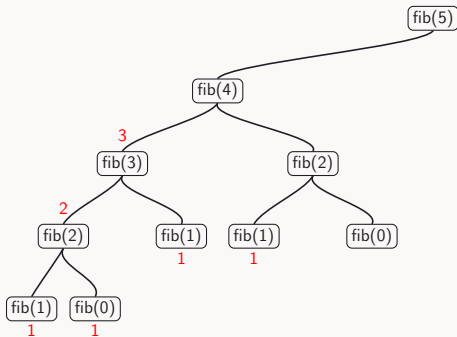
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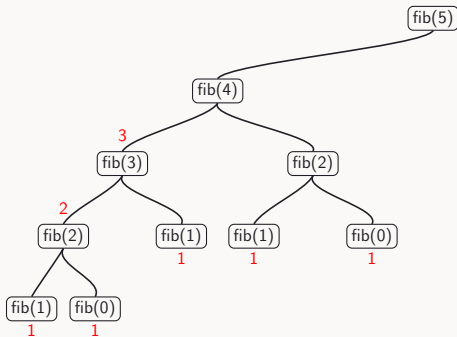
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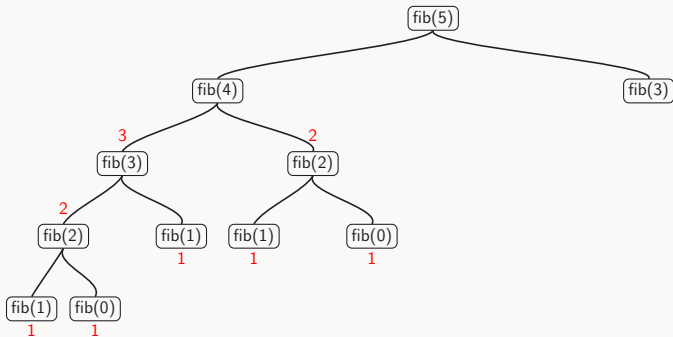
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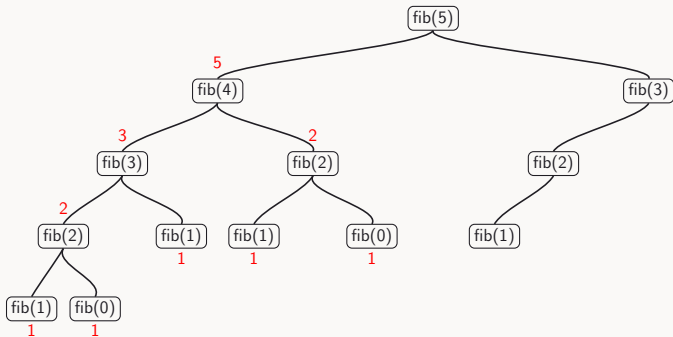






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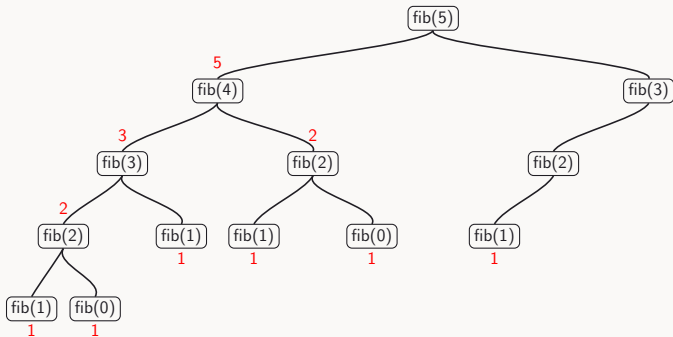
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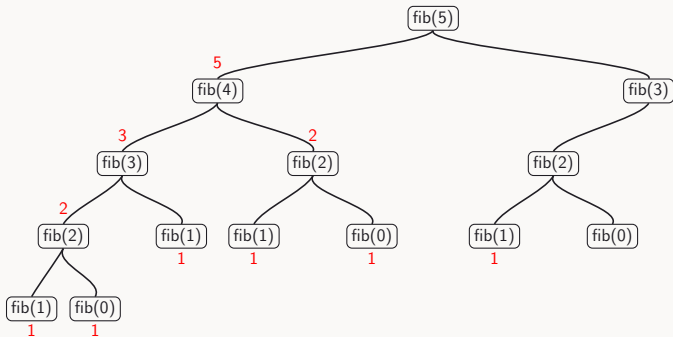
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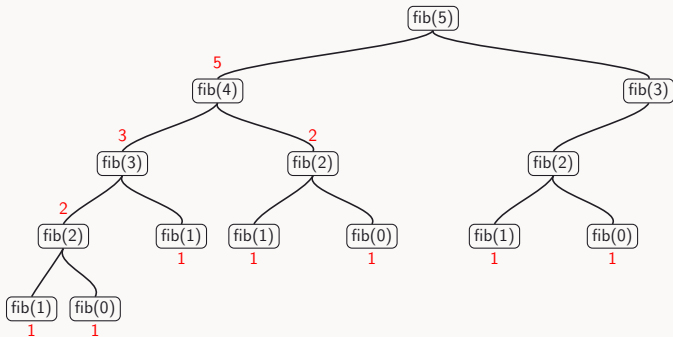
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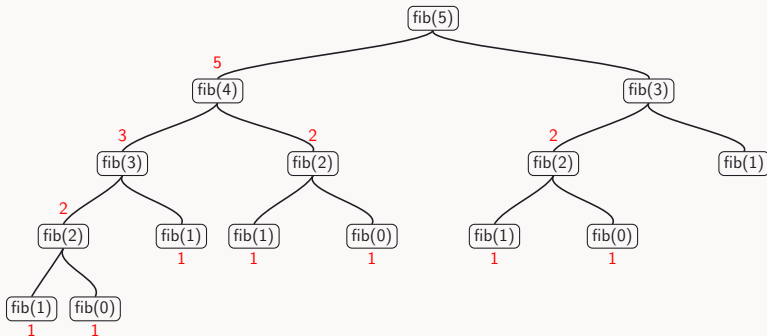
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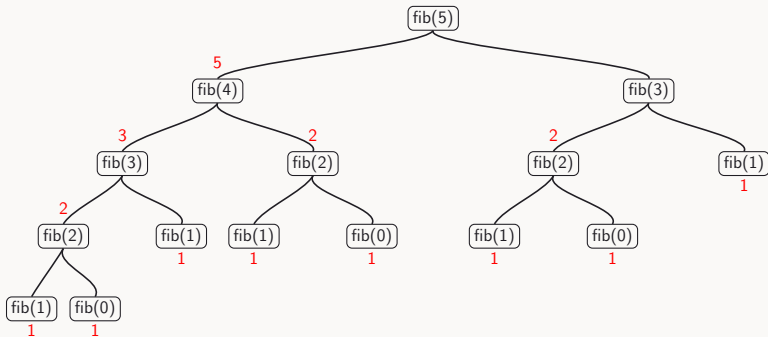
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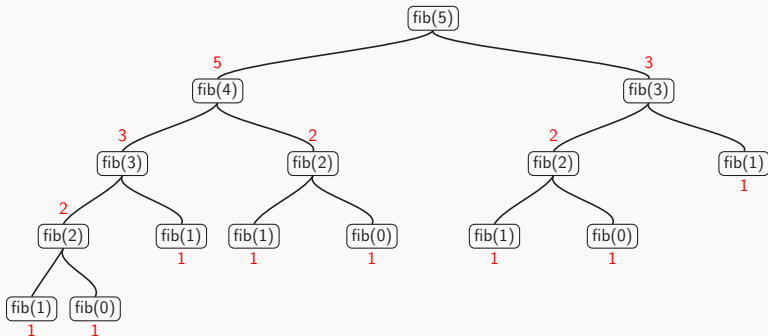
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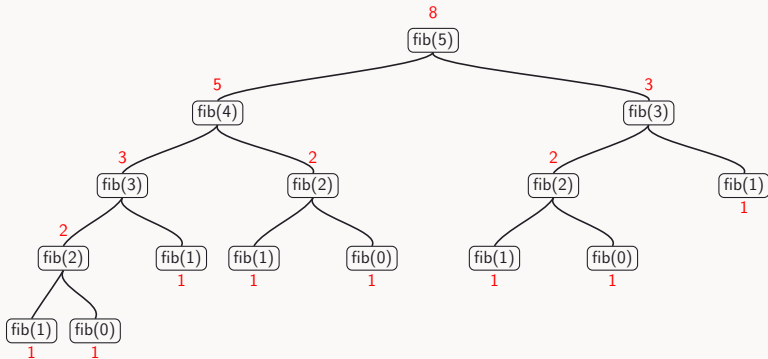
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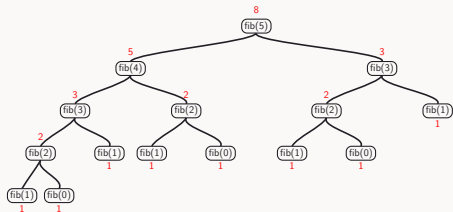
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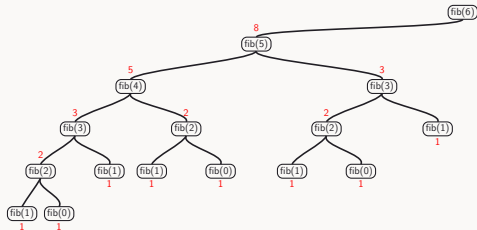


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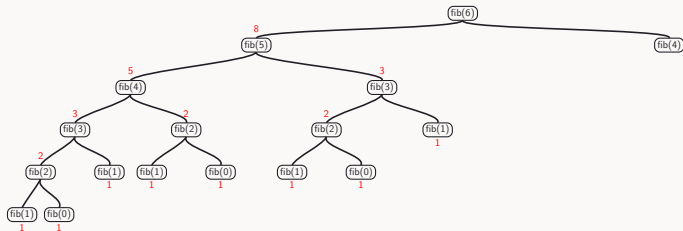


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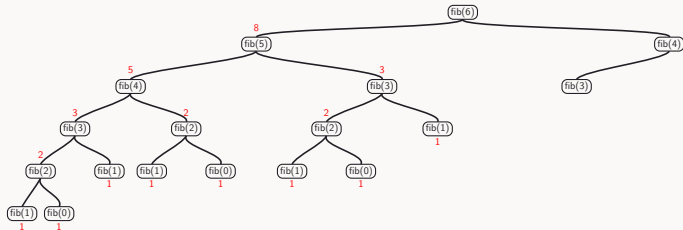


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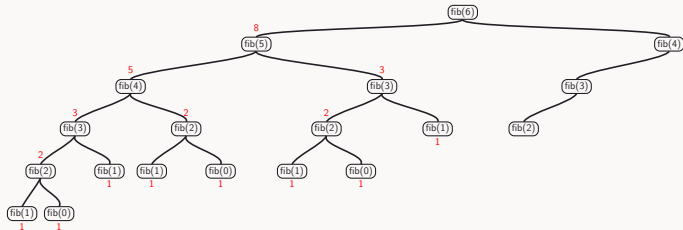


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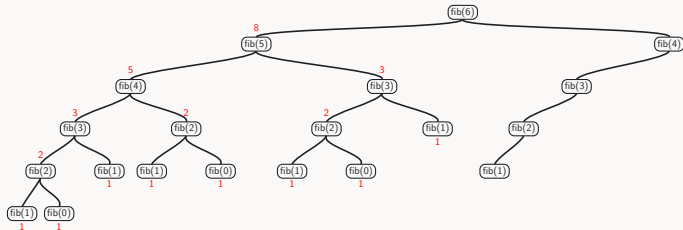


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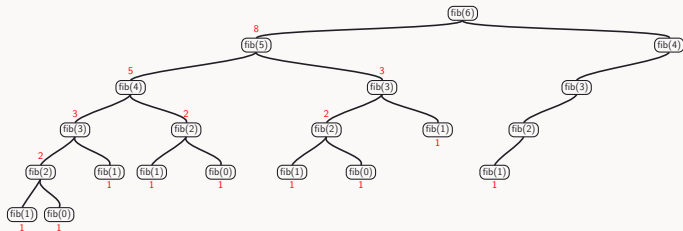


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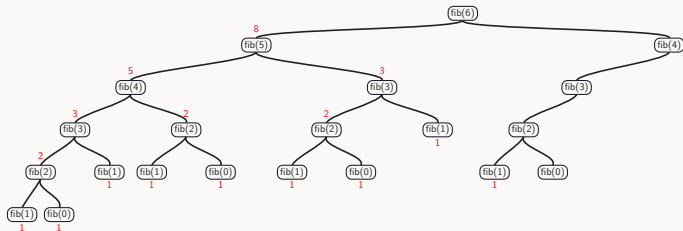


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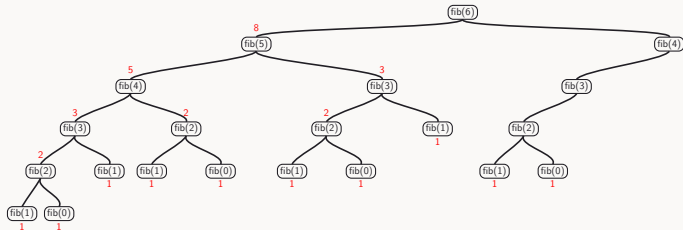


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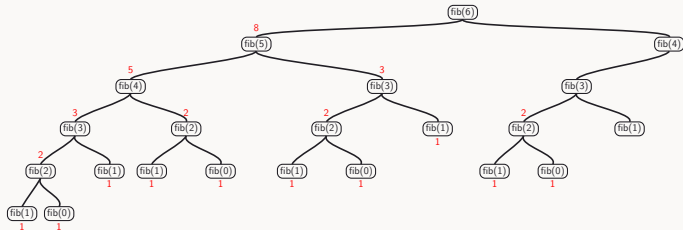


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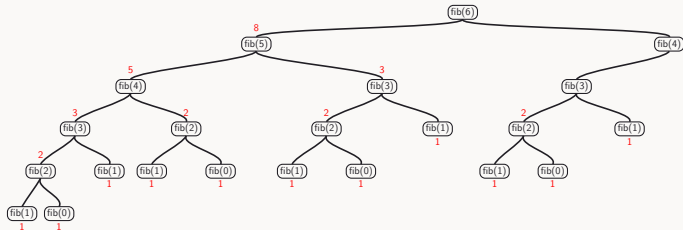


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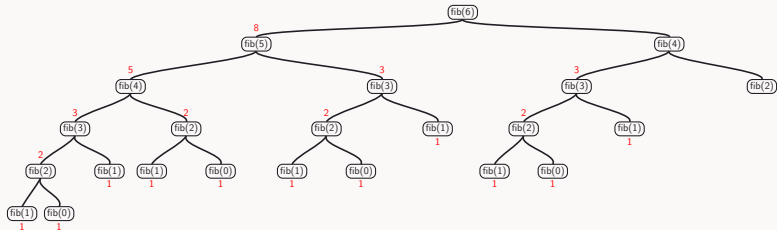


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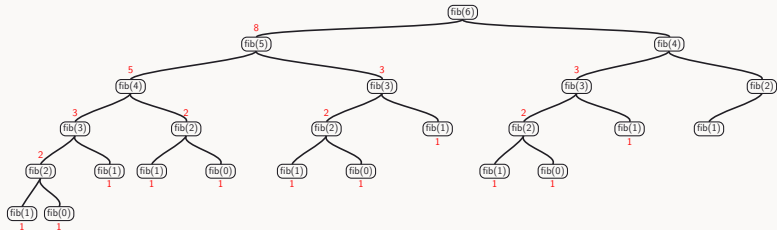


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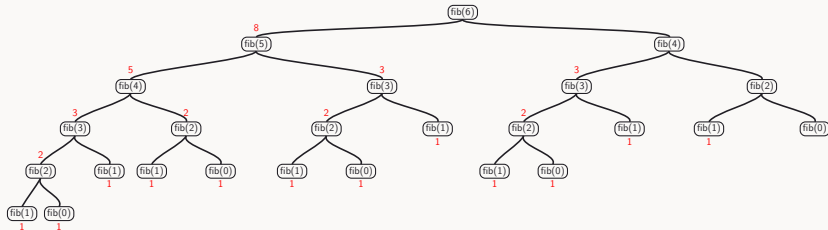
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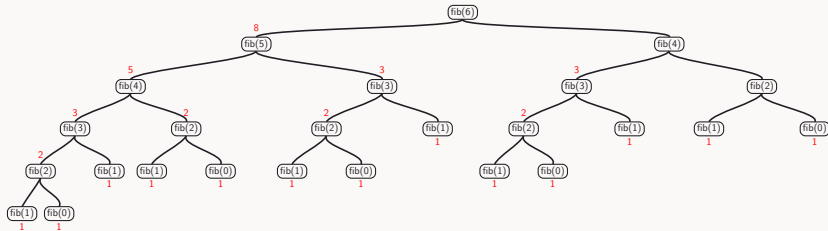




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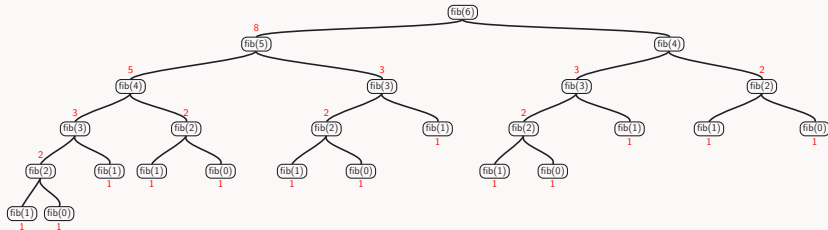


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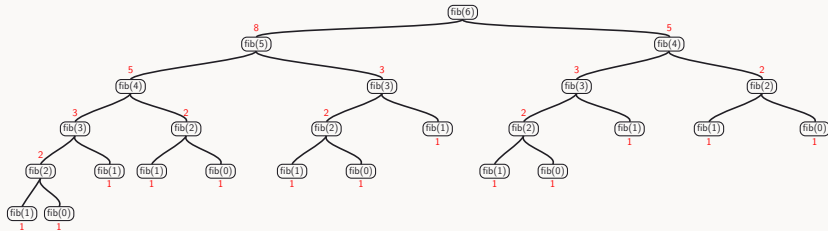




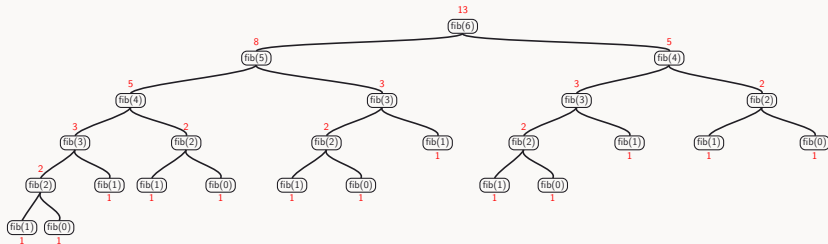
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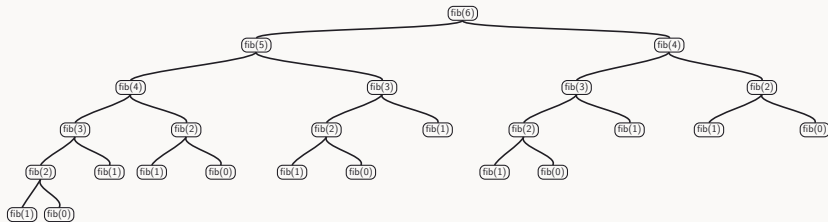


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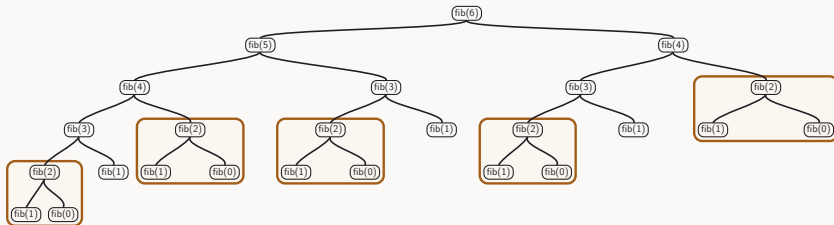


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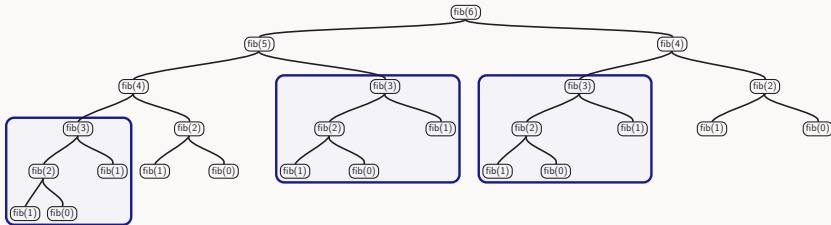




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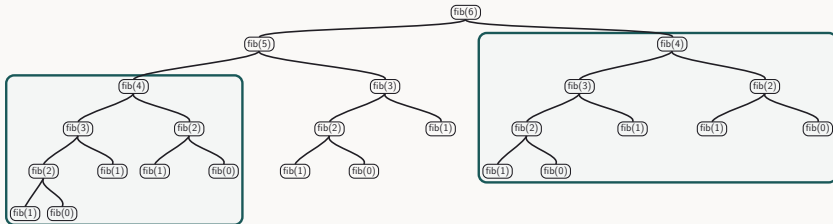


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Example: Computing $F(n)$ using $\text{fib}(n)$

- ▶ Number of leaves in the recursion tree for $\text{fib}(n)$ is $F(n + 1)$
 - $\text{fib}(10)$ needs to visit $F(11) = 89$ leaves
 - $\text{fib}(20)$ needs to visit $F(21) = 10946$ leaves
 - $\text{fib}(40)$ needs to visit $F(41) = 165580141$ leaves
 - $\text{fib}(100)$ needs to visit $F(101) = 573147844013817084101$ leaves



Time for Evaluating $\text{fib}(n)$

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Time for Evaluating fib(n)

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- ▶ Tree for fib(n) has $F(n + 1)$ leaves, where
 - $F(n) = \left\lfloor \frac{\varphi^n}{\sqrt{5}} + \frac{1}{2} \right\rfloor$, and $\varphi = \frac{1 + \sqrt{5}}{2}$

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Can we write an efficient iterative function that computes $F(n)$?



Fibonacci Numbers (iterative version)



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```
def fib(n):  
    return fib_iter(1, 0, n)  
  
def fib_iter(a, b, count):  
    return b if count == 0 else fib_iter(a + b, a, count  
    - 1)
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- How efficient is this program?



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► How efficient is this program?

Wait. What does it even mean to be efficient?



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A Tale of Two Programmers



Prof Martin

writes beautiful, elegant code



Prof Sanka

writes ruthlessly efficient code



A Tale of Two Programmers



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writes beautiful, elegant code



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Both are asked to solve the *same* problem.



How Do We Compare Their Programs?

- ▶ Readability



How Do We Compare Their Programs?

- ▶ Readability
- ▶ Style



How Do We Compare Their Programs?

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- ▶ Style
- ▶ Elegance



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Wait, how do we compare programs?



How Do We Compare Their Programs?

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Wait, how do we compare programs?

Today: two measurable dimensions

Time: how long does the program run

Space: how much memory does it need

Recall the Factorial Function

- ▶ Recursive definition
 - $n! = 1$ if $n = 1$
 - $= n(n-1)!$ if $n > 1$



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def factorial(n):  
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- ▶ Example call:



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- ▶ Example call:

```
print(factorial(4))
```





Time for Calculating factorial(n)

```
factorial(4)
```



Time for Calculating factorial(n)

```
factorial(4)  
4 * factorial(3)
```



Time for Calculating factorial(n)

```
factorial(4)  
4 * factorial(3)  
4 * (3 * factorial(2))
```



Time for Calculating factorial(n)

```
factorial(4)
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4 * (3 * (2 * factorial(1)))
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Time for Calculating factorial(n)

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4 * factorial(3)
4 * (3 * factorial(2))
4 * (3 * (2 * factorial(1)))
4 * (3 * (2 * 1))
```



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```
factorial(4)
4 * factorial(3)
4 * (3 * factorial(2))
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4 * (3 * (2 * 1))
4 * (3 * 2)
```



Time for Calculating factorial(n)

```
factorial(4)
4 * factorial(3)
4 * (3 * factorial(2))
4 * (3 * (2 * factorial(1)))
4 * (3 * (2 * 1))
4 * (3 * 2)
4 * 6
```



Time for Calculating factorial(n)

```
factorial(4)
4 * factorial(3)
4 * (3 * factorial(2))
4 * (3 * (2 * factorial(1)))
4 * (3 * (2 * 1))
4 * (3 * 2)
4 * 6
24
```



Time for Calculating factorial(n): Observation

Observation

Number of operations grows linearly proportional to n



Space for Calculating factorial(n)

```
factorial(4)
```



Space for Calculating factorial(n)

```
factorial(4)  
4 * factorial(3)
```



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```
factorial(4)  
4 * factorial(3)  
4 * (3 * factorial(2))
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factorial(4)
4 * factorial(3)
4 * (3 * factorial(2))
4 * (3 * (2 * factorial(1)))
4 * (3 * (2 * 1))
4 * (3 * 2)
4 * 6
```



Space for Calculating factorial(n)

```
factorial(4)
4 * factorial(3)
4 * (3 * factorial(2))
4 * (3 * (2 * factorial(1)))
4 * (3 * (2 * 1))
4 * (3 * 2)
4 * 6
24
```



Space for Calculating factorial(n): Observation

Observation

Number of deferred operations grows linearly proportional to n



Doubling the Argument of factorial ($n = 2$)

```
factorial(2)
```



Doubling the Argument of factorial ($n = 2$)

```
factorial(2)  
2 * factorial(1)
```




Doubling the Argument of factorial ($n = 2$)

```
factorial(2)  
2 * factorial(1)  
2 * 1
```



Doubling the Argument of factorial ($n = 2$)

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2 * factorial(1)
2 * 1
2
```



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Doubling the Argument of factorial: Observation

Time

The number of steps “roughly” doubles. The factorial function runs in a time (linearly) proportional to the argument n .



Doubling the Argument of factorial: Observation

Time

The number of steps “roughly” doubles. The factorial function runs in a time (linearly) proportional to the argument n .

Space

The number of deferred operations also “roughly” doubles. It has a space requirement (linearly) proportional to the argument n .



Orders of Growth

- ▶ Linear growth
 - The factorial function runs in a time (linearly) proportional to the argument n



Orders of Growth

- ▶ Linear growth
 - The factorial function runs in a time (linearly) proportional to the argument n
- ▶ Exponential growth
 - The fib function runs in a time that grows exponentially with the argument n



Orders of Growth

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- ▶ Exponential growth
 - The fib function runs in a time that grows exponentially with the argument n

What exactly do we mean by the above?



Purpose

- ▶ Rough measure
 - We are interested in a rough measure of resources used by a computational process, with respect to the problem size
- ▶ Abstraction
 - “Order of growth” is an abstraction technique
 - We decide to ignore details that we deem irrelevant
 - Examples: the processor speed of the computer, the programming environment, the programming language, or minor differences in programming style



The Resource Function

- ▶ For a program P solving a problem of size n , let $r(n)$ denote the amount of a *resource* (time or space) that P needs



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The Resource Function

- ▶ For a program P solving a problem of size n , let $r(n)$ denote the amount of a *resource* (time or space) that P needs
- ▶ Recall Prof Martin and Prof Sanka: both write a program for the *same* problem
- ▶ Plug in Martin's program: $r(n)$ tells us its resource use
- ▶ Plug in Sanka's program instead: $r(n)$ now tells us Sanka's program's resource use, same symbol, different program

The Θ (“Big Theta”) Notation

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- ▶ Definition: The function r has order of growth $\Theta(g(n))$ if there are positive constants k_1 and k_2 and a number n_0 such that $k_1 \times g(n) \leq r(n) \leq k_2 \times g(n)$ for all $n > n_0$

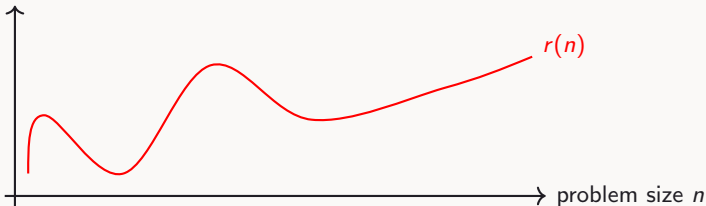
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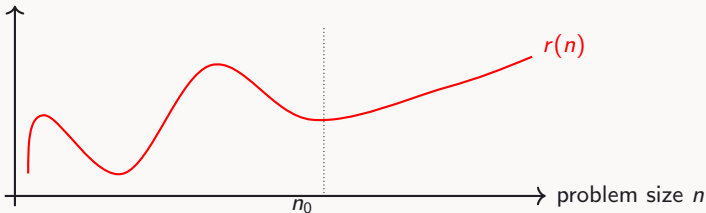
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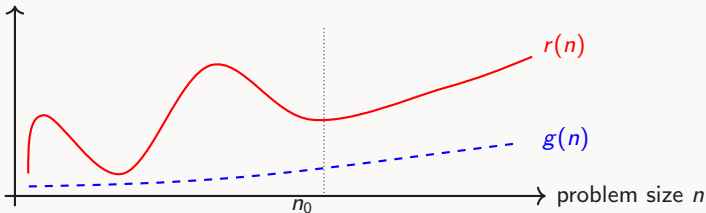
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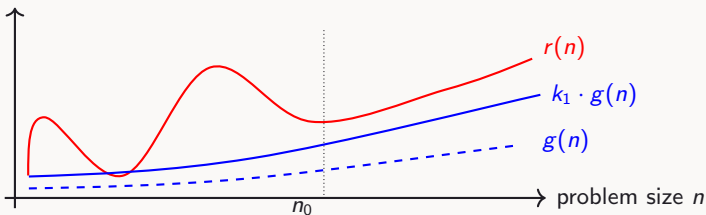
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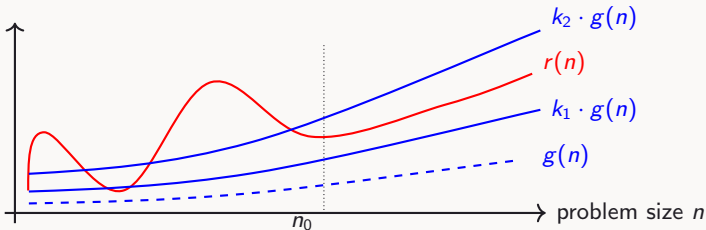
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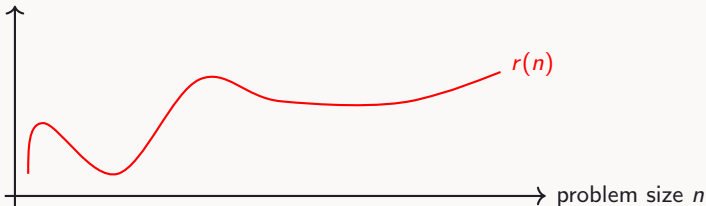
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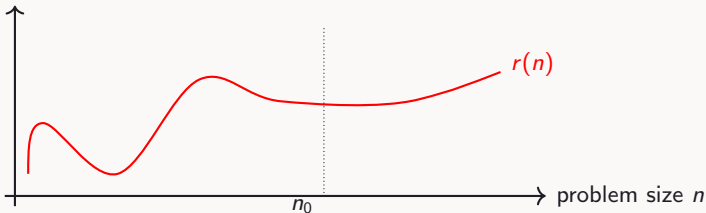
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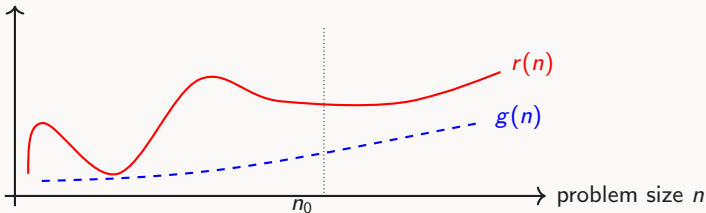
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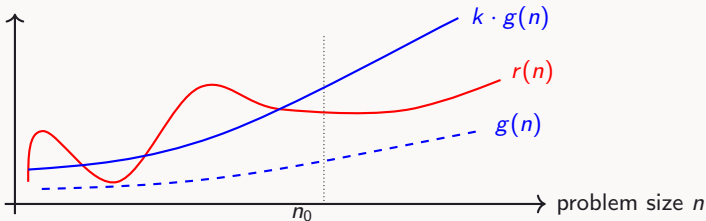
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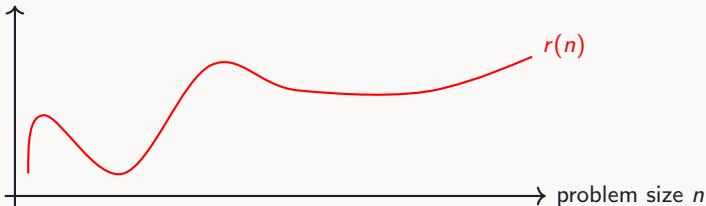
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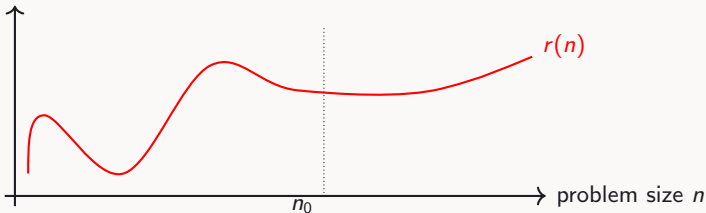
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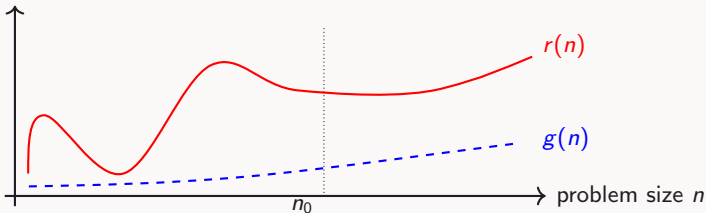
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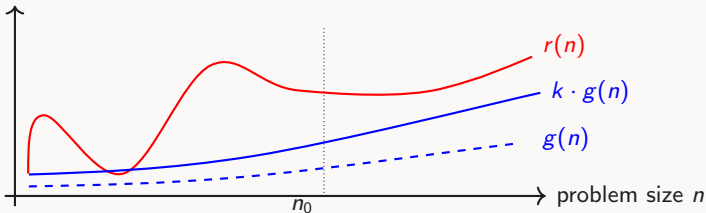
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Do Constants Matter?

- ▶ Let's say r has order of growth $\Theta(n^2)$
- ▶ Does r also have order of growth $\Theta(0.5n^2)$?
- ▶ Constants don't matter
 - Because we can freely choose k, k_1, k_2

Do Minor Terms Matter?

- ▶ Let's say r has order of growth $O(n^2)$

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- ▶ Let's say r has order of growth $O(n^2)$
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Do Minor Terms Matter?

- ▶ Let's say r has order of growth $O(n^2)$
- ▶ Does r also have order of growth $O(n^2 + 40n - 5)$?
- ▶ Minor terms don't matter
 - Because we can adjust n_0 , k , k_1 , k_2 such that the minor terms are overruled



Some Common $g(n)$

► $\log n$



Some Common $g(n)$

- ▶ $\log n$
- ▶ n



Some Common $g(n)$

- ▶ $\log n$
- ▶ n
- ▶ $n \log n$



Some Common $g(n)$

- ▶ $\log n$
- ▶ n
- ▶ $n \log n$
- ▶ n^2



Some Common $g(n)$

- ▶ $\log n$
- ▶ n
- ▶ $n \log n$
- ▶ n^2
- ▶ n^3



Some Common $g(n)$

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- ▶ n
- ▶ $n \log n$
- ▶ n^2
- ▶ n^3
- ▶ 2^n



Growing Pains



Growing Pains

It is now time (or space) to go to the Source Academy!

Apologies, I am running out of jokes here already and we are only in week 2!



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(No need to finalize your Check-In submission. Please stop working as soon as we call time!)



Example Orders of Growth

- ▶ Linear time
 - The `factorial` function runs in a time that has order of growth $\Theta(n)$, where n is the function argument



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► Exponential time

- The `fib` function runs in a time that has order of growth $\Theta(\varphi^n)$, where n is the function argument and $\varphi = \frac{1+\sqrt{5}}{2} \approx 1.618$



Example Orders of Growth

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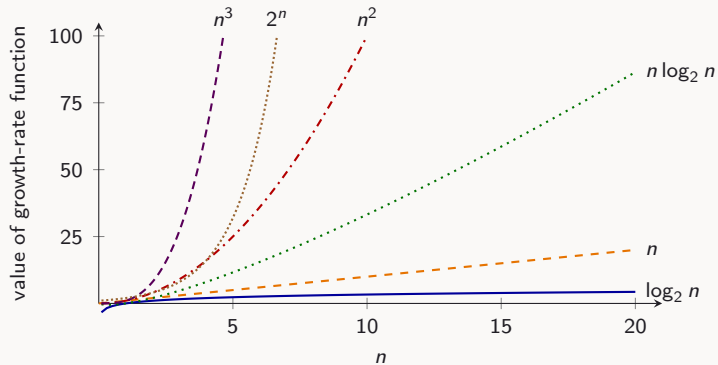
► Exponential time

- The `fib` function runs in a time that has order of growth $\Theta(\varphi^n)$, where n is the function argument and $\varphi = \frac{1+\sqrt{5}}{2} \approx 1.618$
- Can we say its order of growth is $O(2^n)$?

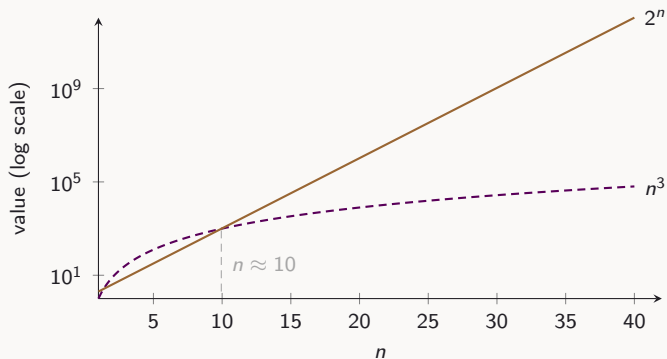
Comparing Orders of Growth: Table

Function	n					
	10	100	1,000	10,000	100,000	1,000,000
1	1	1	1	1	1	1
$\log_2 n$	3	6	9	13	16	19
n	10	10^2	10^3	10^4	10^5	10^6
$n \log_2 n$	30	664	9,965	10^5	10^6	10^7
n^2	10^2	10^4	10^6	10^8	10^{10}	10^{12}
n^3	10^3	10^6	10^9	10^{12}	10^{15}	10^{18}
2^n	10^3	10^{30}	10^{301}	$10^{3,010}$	$10^{30,103}$	$10^{301,030}$

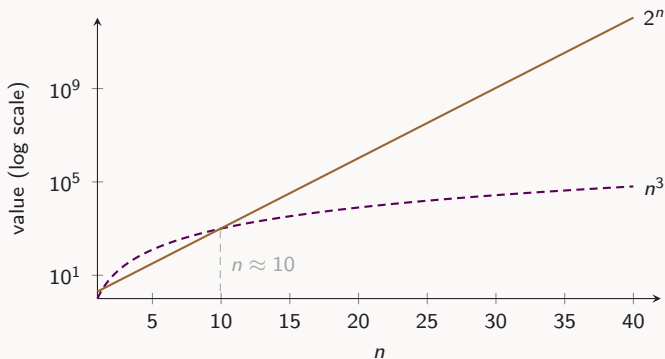
Comparing Orders of Growth: Graph



Comparing Orders of Growth: n^3 vs 2^n

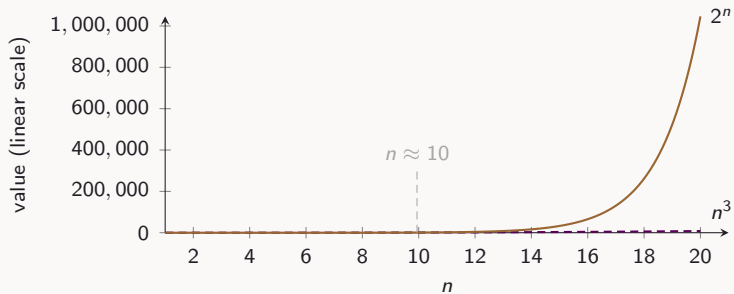


Comparing Orders of Growth: n^3 vs 2^n

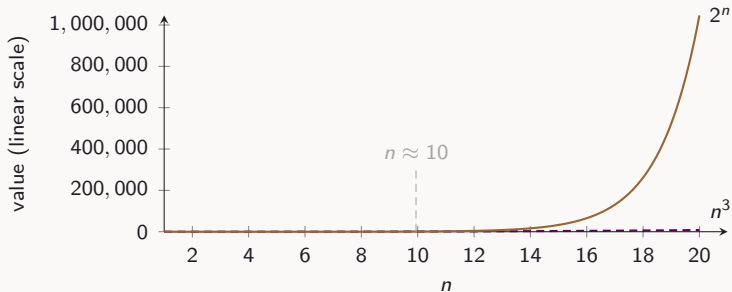


They cross around $n \approx 10$ and past that, 2^n pulls away from n^3 ever faster, and the gap grows without bound.

n^3 vs 2^n : Linear Scale



n^3 vs 2^n : Linear Scale



On a linear scale, n^3 all but disappears next to 2^n once n grows a little past the crossover this is exactly why we use order of growth instead of trying to compare raw numbers.



How to Calculate “Big Oh/Theta/Omega”

- ▶ Topic of Algorithm Analysis (CS3230)
- ▶ For us:
 - Identify the basic computational steps
 - Try a few small values
 - Extrapolate
 - Watch out for “worst case” scenarios



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Why tightest?

It's true that Usain Bolt can run 100m in under 10 *minutes*... but that's not meaningful. What we actually care about is that he can do it in under 10 *seconds*.

Summary

- ▶ Resources for computational processes: time and space
- ▶ Big Theta, Big Oh, Big Omega
 - Provide abstractions or “rough” measures of resources used with respect to problem size